

Gephi Network Analysis Report: Amazon Sampled Graph

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1. Introduction

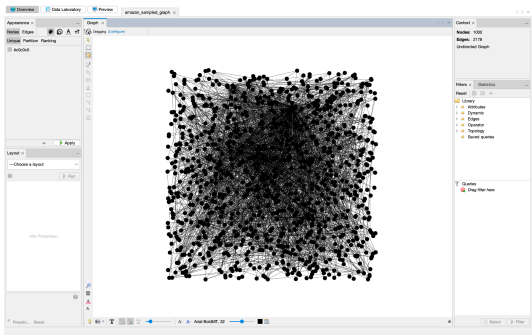
This report presents a network analysis of the **Amazon sampled graph** using **Gephi**. The dataset was imported as an **undirected graph**. In this network, nodes represent entities in the dataset, and edges represent relationships or connections between them.

2. Dataset Import

The CSV file was imported into Gephi as an **undirected graph**. After importing, the network contained:

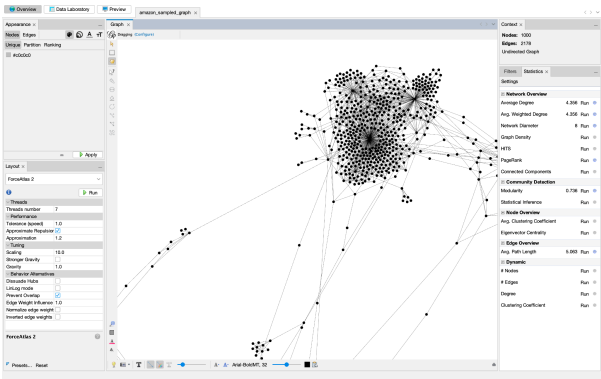
Network Property	Value
Number of nodes	1,000
Number of edges	2,178
Graph type	Undirected
Average degree	4.356
Average weighted degree	4.356
Network diameter	8
Average path length	5.063

The graph was successfully loaded into the **Overview** workspace of Gephi.



3. Graph Layout and Visualization

To improve the readability of the network, the **ForceAtlas2** layout was applied. After applying ForceAtlas2, the network became more readable. Dense clusters moved closer together, while smaller groups and bridge structures became easier to see.



4. Degree Analysis

Degree measures how many direct connections a node has. A node with a high degree is directly connected to many other nodes and can be considered a **hub** in the network. The top 5 nodes with the highest degree are:

Rank	Node ID	Degree
1	458358	324
2	38365	151
3	175102	92
4	498702	46
5	37981	46

The most connected node is **458358**, with a degree of **324**. This means it has far more direct connections than any other node in the network. Node **38365** is the second strongest hub, with **151** connections.

These high-degree nodes are important because they can represent popular or highly connected entities in the Amazon network.

5. Betweenness Centrality

Betweenness Centrality measures how often a node lies on the shortest path between other nodes. A node with high betweenness can act as a **bridge** between different parts of the network. The top nodes with high betweenness centrality are:

Rank	Node ID	Betweenness Centrality
1	458358	235624.81744
2	24139	212579.94642
3	38365	101842.66447
4	175102	48184.63559
5	222939	24306.01794

Node **458358** has the highest betweenness centrality. This means it is not only highly connected, but also important for connecting different areas of the network. Node **24139** is also very important, even though it does not have as many direct connections as the highest-degree nodes. Its high betweenness value suggests that it works as a bridge between different communities.

Id	Label	Interval	Degree	Weighted Degree	Eccentricity	Closeness Centrality	Harmonic Closeness Centrality	Betweenness Centrality	PageRank	Modularity Class
161707		2	2.0	8.0	0.007659	0.037617	186.207254	0.000028	0	
36000		15	15.0	8.0	0.236225	0.2361	2382.88607	0.003191	1	
101221		41	41.0	7.0	0.244374	0.29022	27510.99024	0.009132	0	
272396		4	4.0	8.0	0.259559	0.277918	1.0	0.000064	1	
38365		101	101.0	7.0	0.260914	0.437306	101842.664473	0.033413	1	
470943		4	4.0	8.0	0.259559	0.277918	0.333333	0.000002	1	
477105		4	4.0	8.0	0.259559	0.277918	0.333333	0.000002	1	
913207		3	3.0	8.0	0.255548	0.277416	0.0	0.000791	1	
126037		1	1.0	8.0	0.254546	0.276415	0.0	0.000339	1	
366510		6	6.0	7.0	0.163034	0.160542	4970.3	0.001156	2	
56488		1	1.0	7.0	0.154668	0.158078	0.0	0.000405	2	
264861		2	2.0	7.0	0.154668	0.158078	0.0	0.000716	2	
377917		1	1.0	7.0	0.154668	0.158078	0.0	0.000405	2	
470072		7	7.0	5.0	0.22334	0.238186	5962.3	0.001426	10	
481937		1	1.0	7.0	0.154668	0.158078	0.0	0.000405	2	
483741		2	2.0	7.0	0.154668	0.158078	0.0	0.000792	2	
266365		4	4.0	8.0	0.247192	0.333726	154.044172	0.000071	6	
175102		82	82.0	7.0	0.277731	0.384346	48184.635587	0.019378	3	
431174		8	8.0	8.0	0.244913	0.332554	186.189703	0.001648	5	
468358		324	324.0	7.0	0.306358	0.519214	235624.81744	0.069371	5	
516125		4	4.0	7.0	0.240789	0.284047	1490.0	0.001108	3	
261786		17	17.0	8.0	0.168811	0.280203	215275.956415	0.000869	3	
60917		2	2.0	8.0	0.188057	0.217401	0.666667	0.000007	3	
427363		1	1.0	8.0	0.187979	0.216733	0.0	0.000086	3	
383004		4	4.0	8.0	0.23945	0.306426	1.333333	0.000009	5	
225526		3	3.0	8.0	0.259542	0.306426	1.110487	0.000793	5	
264900		3	3.0	8.0	0.259384	0.306026	0.333333	0.000791	5	
377196		3	3.0	8.0	0.259384	0.306026	0.333333	0.000791	5	
150787		3	3.0	7.0	0.188723	0.172002	2.286581	0.000644	4	
175649		6	6.0	7.0	0.18822	0.172018	2.188657	0.001108	4	
225539		32	32.0	8.0	0.187747	0.217501	24306.017944	0.000059	4	
481404		11	11.0	7.0	0.163636	0.183359	509.821782	0.001999	4	

6. PageRank

PageRank measures influence based not only on the number of connections, but also on the importance of the nodes connected to it. The top 5 nodes by PageRank are:

Rank	Node ID	PageRank
1	458358	0.069371
2	38365	0.033413
3	175102	0.019378
4	521221	0.009512
5	498702	0.008820

Node **458358** again ranks first. This confirms that it is the most influential node in the network according to several measures: degree, betweenness centrality, and PageRank. Node **38365** and node **175102** are also highly influential because they have both high degree and high PageRank.

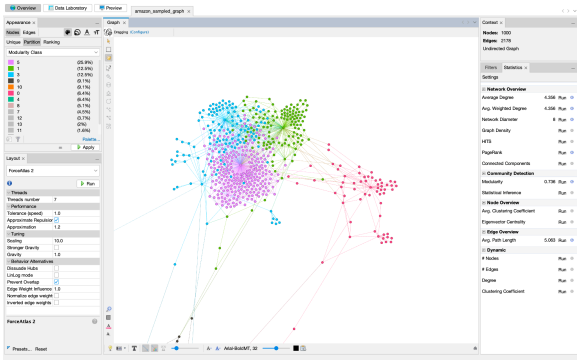
7. Community Detection

Community detection was performed using Gephi’s **Modularity** algorithm. This method groups nodes into communities based on how strongly they are connected to one another. The modularity score was 0.736.

A modularity score of **0.736** indicates that the network has a clear community structure. This means that nodes are not randomly connected. Instead, they form several groups or clusters. After running modularity, nodes were colored according to their **Modularity Class**. Each color represents a different community. The largest visible communities include:

Modularity Class	Approximate Share
5	25.9%
1	12.5%
3	12.5%
9	9.1%
10	9.1%
0	6.4%
4	6.4%

The largest community is **Modularity Class 5**, containing about **25.9%** of all nodes.



8. Filtering and Highlighting

Filtering was used to focus on the most important nodes in the network.

8.1 High-Betweenness Nodes

Nodes with the highest betweenness centrality were identified because they act as bridges between communities. Important bridge nodes include:

Node ID	Role
458358	Main hub and strongest bridge
24139	Major bridge between communities
38365	Strong hub and connector
175102	Important central node
222939	Smaller but relevant bridge node

These nodes are important because removing them could make the network less connected or separate some communities from each other.

8.2 High-Degree Nodes

Nodes with high degree were isolated using a degree filter. These nodes are important because they have many direct connections. The main high-degree nodes are:

Node ID	Degree
458358	324
38365	151
175102	92
498702	46
37981	46

These nodes should be made larger in the final visualization. The top 5 degree nodes should also be colored **black** to make them stand out clearly from the community colors.

9. Interpretation

The network shows a clear clustered structure. The modularity score of **0.736** confirms that nodes are organized into distinct communities.

Node **458358** is the most important node in the network. It has the highest degree, highest betweenness centrality, and highest PageRank. This means it is both a major hub and a major bridge.

Nodes **38365** and **175102** are also highly important because they have high degree and high PageRank. They are strongly connected and influential inside the network.

Node **24139** is especially interesting because it has high betweenness centrality. This means it may not be the biggest hub, but it plays an important role in connecting different communities.

Overall, the network contains several strong communities, with a few highly influential nodes connecting the structure together.

10. Conclusion

The Gephi analysis successfully identified the main hubs, influential nodes, and communities in the Amazon sampled graph.

The most important finding is that node **458358** dominates the network. It is the strongest node according to degree, betweenness centrality, and PageRank. This makes it the most influential node in the graph.

Community detection also showed that the network is divided into several meaningful groups. These communities may represent groups of closely related products or entities.